

Comm 494 Research Project Report

Rishabh Narayan

I applied for supervised undergraduate research this semester with Professor Sundar, as a way to broaden my market research knowledge as well as learn more about AI integration within media for early development of my thesis. During the semester I had the ability to participate in three different experiences ranging from direct application of LLMs in everyday research (Deepfake proj.), literature review, study framing and survey pretesting (Deepfake pre./one-off) as well as feedback/review within ideation stages and development (Misc. from lab group meetings) along with 2 developed projects with feedback from Eunhae and Jack. While I had participated in research experiences before through the psychology side within social mobility and neuro-processing with simple Qualtrics sampling and SONA participant running, this experience has allowed me to learn significantly more about the newly paved territory of AI integration within various media content types through tasks I had not yet had much experience in prior, as well as an understanding on how this can further develop within the world of media strategy and advertising in the future, a field I am very interested in pursuing after my undergraduate coursework. A simple rundown of the tasks I participated in are presented below:

- Study parameter framing/restructuring for the Deepfake Study Pretest
- Survey review/pretesting occasionally for one-off studies
- Two full scale Deepfake products for testing within studies
- Media analysis, Prompt Engineering, and Priming for Deepfake Study Project
- Feedback and review throughout several studies
- Literature review of previous AI integration software
- Group meetings with the Deepfake group + small circle testing with Jack and

Eunchae as well as weekly lab group meetings.

I will reflect on what I have learned from this experience in more detail as follows.

My research experience consisted of study framing, literature and media review, meeting planning, and LLM based application through project development in a wide array of tasks within my three experiences with two of three taking primary focus on the Deepfake pretesting and study framing along with Deepfake project construction with Eunchae and Jack as well as contributorial roles within survey pre-testing and course review. A courseload of 15 weeks and around 135+ hours were spent collaborating with the lab group and working with research studies and project development. My major tasks when assisting with study parameters and literature review within these experiences included taking notes from past studies and media to summarize its value points for future project use, designing and developing LLM deepfake video creation through Pinokio, Seedance, and Higgsfield that would not utilize more than 16 GBs of storage and bypass political filters, ran circle testing meetings (2) with Jack and Eunchae with literature/media findings within LLM as well as ran a tutorial on how to set up and create their own, priming multiple video generations through prompting, faceswap, and speed modulation, explaining study procedure and presenting script edits to a small group, presenting feedback and testing suggestions regarding various projects, and analyzing past data trends to make a synopsis. This experience helped me understand more about project design testing in many aspects as follows. It made me aware that sometimes your first output won't be one that will succeed immediately, despite the large amount of research done to ensure its quality. Sometimes it takes a 5th trial, a 10th or even 100th to truly understand even the fundamentals of the design. I remember when I initially started

researching deepfakes and presenting information to our team, I thought I would be able to immediately run tests because I had done deepfakes in the past as well as did research to get to a point of development, however once I actually followed what I had learned and picked a model that was supposedly “the best on the market” I ended up with generations that were extremely messed up, choppy, or fast in speech furthermore there were rarely any tutorials or studies out there that actually showed me how to develop a modern AI deepfake integration tool after the development of Seedance 2.0, which led to a lot of trial and experimentation through LTX design tools, part swapping within Pinochiko, and finally figuring out the sweet spot regarding Higgsfield AI and physical video editing without the use of AI in order to reduce choppiness and seamless integration. Despite what I had already experienced before in prior deepfakes, which were simple overlap tools done in Google Drive, the game has changed in the past 3 years and I had to adapt to the new tools and push through and learn from these errors to finally make the fully developed product used in the study in the end. Regarding the tasks that I had completed during this project, I wanted to use this next section as a way to go into depth regarding how I formulated, addressed drawbacks, and developed my final work within these three experiences. Prior to Deepfake project creation, I was heavily involved within the pretesting side regarding study framing and goal curation. To choose the best candidate for our model, we analyzed several behavioral aspects such as audience influentialness, personal values, and moral attunement as well as adjusted parameters regarding data collection for our future subject pool. I.e Gen-Z most susceptible to digital media. On top of simply modeling and creating the deepfake for the Deepfake study I also assisted with developing the script, researching/establishing literature review on prior deepfake applications, as well as presenting this data within our small circle group

with a presentation and live tutorial. When scripting the original two deepfake scenarios that I had worked on, we had to figure out a way to keep the authenticity of the initial reference but also keep it concise enough to make it within a 15 second limit (AI capabilities would only allow for 15 second video generation due to an overload on GB). While the initial generations were frustrating and required a lot of priming to get the feeling of authenticity (over 30 generations, and about 2 hours of background generation per day), I was eventually able to develop a script by segmenting the main points into three parts that could seamlessly be edited together as if nothing truly happened. I achieved this effect by splitting clips, remodulating voice speed to overlap with mouth movements, as well as removing and recreating new backgrounds through a green screen effect, all edited through Adobe express after initial generation. On the literature review side for the Deepfake study pretesting & project I was given the chance to learn about unique concepts and past experiments that correlated with our findings, through this I was able to utilize my skills from prior experience such as my social mobility lab to segment and find open material that related to our topics. By completing this literature review within the Deepfake study pretesting and Deepfake project I was able to sharpen my skills of media analysis and prior data functionality and translate that into development outputs. I also had a chance to meet with Eunhae and Jack to present my initial research on deepfake applications, as well as periodic presentations regarding what I had discovered through review/what I was working on currently within the project. Through these meetings I had learned a lot about how to present findings from initial research and how to shape this format into a more digestible content splice through slides, something that I'm sure will carry me in future research instances as I develop my thesis. I was also able to give an in-depth tutorial on how to create deepfakes through

specific LLMs as well as how to set up simultaneous processing through the lab computers. This felt like a full circle moment as I was able to give back to the grad students that had helped me start the initial research in the first place. The first round of results in the lab group meeting sharpened my skills of utilizing data analyzing tools and reading the result outputs. Finally, I was given the opportunity to collaborate, talk through, and give feedback on a variety of projects throughout the semester outside of lab meetings. Grad members like Davis, allowed me to discuss with them on certain projects as well as how specific trials or variations within these studies could actually facilitate more fluid responses and understanding across not just higher level researchers, but college students and high schoolers as well. I learned just how important cross referencing can be when running studies that involve members of less experience within the communication or AI space, as misunderstanding words or phrasing can completely skewer results as well as the quality of answer within these represented data sets. Especially if these results are facilitated through school work, job application review, or any common experience that would generate such a large data pool. Overall, I am very happy with how this final project turned out regarding all these experiences. I was able to learn so much about deepfake models through this experience, as well as how to approach it in a later study if it ever pops up again. I also was able to learn a lot about how to collect information from a large pool of sample articles/media and translate that into a full working demo and presentation. Furthermore, presenting the findings in the official lab group meeting towards the end of the semester was also very insightful, because it allowed me to see a general synopsis and feedback scale that was outside of a metric that was only my own up till that point. This allowed me to refine the current LLM model and directly focus on cleaning up the specific metrics we talked about (speed, timing, etc) so that it was

ready for Jack as he took it over for the summer.

Last but not least, I participated in the lab group meeting every Friday which was always a unique experience. Having only participated in psychology based research before, getting to hear about specific topics and research that actually related to my future goals within advertising from researchers actually working in the field was eye-opening and truly helped me envision career goals for the future. I was really excited to learn about how media effects play a role in our current media and campaigns, especially within this new age of AI and whether or not it will truly be able to perceive like humans do or be stuck in a loop. I love how the group is very collaborative and open to new ideas, if a suggestion is not up to par it is still acknowledged and debated on. I was also happy to sometimes have the ability to “hold my own” and offer feedback and testing choices when it came to these experiments and designs from all these talented researchers especially if it popped up in future reviews. This collaborative energy is extremely valuable within this space today, as it keeps editing and thinking of new ideas fun without having to feel like going back to ground zero once again.

Overall, the research experience during this semester has been extremely helpful. I have participated in the whole process of conducting a research project, and my research skills (study design, literature review, study reframing, media and content analysis, and presentation) have prepared me thoroughly for a future thesis and hopefully graduate school in the near future which was the initial goal of doing this research project. I really appreciate this opportunity to work with Professor Sundar, and the level of insight I had received from his thinking within communicative efforts at such a high level. Because of this project, future opportunities such as The Hollywood Program and The Schreyer Honors College became much more

achievable. I also appreciate both Jack and Eunhae's help as they always allowed me to ask questions no matter how simple as well as guided me to finding the right resources no matter what project we were currently working on. Thanks to this experience, especially what I have learned within literature review with deepfake and LLM technology as well as media analysis regarding these tools, I now hope to pivot towards helping out during the summer as well as to a thesis of color theory and branding within advertising campaigns and how AI is able to perceive these attributes.

Project 1 and 2 Deepfake Scripting (Final demos available on SLG):

Note.

(1) Face close-up (upper body);

(2) tones and verbal tics of this person if any; e.g., Talarico has a specific rhythmic triplet style (e.g., "Our economy is broken. Our politics are broken. Our relationships are broken."), and this is applied to both scripts. Not sure if he uses any specific tones for speech (so please check!).

1. Corruption & Democracy (opposite of his position)

Original claim: Never taken a dime of corporate PAC money

Deepfake script:

Let's have a moment of real honesty. For years, I've stood in pulpits and on porches promising that we could fix a broken system without touching a single dime of corporate PAC money. I spoke about integrity, I spoke about sacred trust, and I spoke about flipping the tables of the money-changers. But I was wrong. I was wrong about the math, and I was wrong about how change actually happens.

Pretending that corporate influence doesn't exist hasn't lowered your taxes, it hasn't saved our schools, and it hasn't put food on your table. So, I'm changing the game. Starting today, I am accepting every corporate PAC dollar available. We can sit on the sidelines clutching our 'clean' campaigns and losing, or we can step into the halls of power and actually rule. I choose power. I'm not playing the teacher anymore—I'm playing to win.

2. Immigration (more extreme version of his position)

Original claim: We can both welcome the stranger — refugees, asylum seekers, and folks who want to contribute to our economy and pursue the American dream

Deepfake script:

Let's tell the truth: the borders we've built are an affront to the neighbors we claim to love. Immigrants aren't just a part of Texas—they are Texas, and they are the only reason this economy hasn't already collapsed under its own greed.

We've spent billions on raids, walls, and barbed wire, but these aren't security measures—they are tools of state-sponsored terror against the hands that pick our food, build our homes, and save our lives. I am done with the 'legal status' debate. We should stop pretending that a piece of paper makes one human being more 'legal.' It is time to tear down the walls, abolish the checkpoints, and open every door. Texas was built by those who crossed oceans and deserts with nothing, and it belongs to them—not to the politicians in Austin who hide behind a badge. I will fight until every border is a bridge.

Example of scripting work, media analysis, and original deepfake research:

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Narayan, Rishabh K

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To: 📧 Jang, Eunghae

Tue 5/19/2026 8:00 AM

📎 Original James Deepfake Slo...
8 MB

📎 Talurico Successful Deepfake ...
2 MB

📎 Talurico Successful Deepfake ...
2 MB

📄 Show all 6 attachments (17 MB) 📁 Save all to OneDrive - The Pennsylvania State University ⬇ Download all

Hi there, **Eunghae!**

Just wanted to send some updates regarding the deepfakes.

I tried my best to edit the video, but I just don't have the tools to do frame by frame graphing for the background video. The AI model moves around too much, and it doesn't match evenly across all frames, especially if the audio needs to sync up. I instead was able to slow down the video without warping the sound to the best that I could, however the inflections don't match the same energy as the initial. It is the one titled "Original Deepfake Slowed." Good news however is the credits did replenish so I did go ahead and create videos for the second scenario and attached them here, I was able to make the background the same for all videos and the speech and dialogue matches up much more to Talurico's actions.

Some notes for future deepfakes, Higgsfield has a 15 second max for videos (as do every other LLMS at the moment) so the best course of action would be to make a speech/create an action that can be accomplished within the 15 second timeframe for the most believable results and lead to less meshing overall, that way we can use the rest of the credits to fine tuning the background, expressions, and tone inflection. The AI also has a ton of issues with keeping everything uniform with every generation, so I have attached the messed-up generations to hopefully use to train AI in the future as well.

Let me know if you have any questions and am excited to see you guys at the meeting later today!

Rishabh Narayan

For sure, I could definitely figure out something regarding these two videos. I have found two programs that I think would work amazingly for this experiment (attached examples below) Aniface (hero-1) Higgsfield (<https://www.youtube.com/watch?v=e0AkvoA3Yjc>) (Timestamp 3:56). Aniface is around \$13 for the first month and then \$20 every other. You can also pay a one-time purchase of \$20 to get 100 credits to play around with, without a commitment to the subscription. Higgsfield is priced around \$15 every month, but you are sacrificing some features. They had a test trial and I was able to generate a quick 5 second video of James Talarico synced with a trump speech which I have attached below (Used this video as a reference https://www.youtube.com/shorts/QrCERYRb_ZE), you can see that the face tracking is much more advanced than what was used before so overall a much better step in the right direction, no matter what tool we use. (NOTE: we may have to have one of us record a video of us saying this script with motion tracking in order to get the best result since AI uses high quality face tracking the best to recreate, just wanted to put that out there.) Let me know what you think and what you think would be best!

Kind regards,

Rishabh Narayan

- James Deepfake: ▶ [hf_20260407_043153_37cdadb4-5977-4560-a7b7-27218aaed32f.mp4](#)